

The empirical recurrence for OEIS A282373: recovering a conjecture that is not written down

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Abstract

OEIS A282373 records “Empirical recurrence of order 54”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 106$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282373 is “Number of $n \times 4$ 0..1 arrays with no 1 equal to more than four of its king-move neighbors, with the exception of exactly two elements.”. It has offset 1 and begins

0, 6, 309, 6244, 119390, 2131096, 36258678, 594236950, ...

The entry states:

Empirical recurrence of order 54 (see link above)

As of the “Last modified” line on the live entry (Feb 13 2017 11:07:56, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 769$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 106$ of the 769, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 106$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 212$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	27	c_{15}	−22358461462	c_{29}	−63013020285468	c_{43}	−308907669336
c_2	−114	c_{16}	−113443174332	c_{30}	−27500409232384	c_{44}	183026898672
c_3	−894	c_{17}	−128600569374	c_{31}	42628459826052	c_{45}	138792106744
c_4	−8787	c_{18}	204683938272	c_{32}	70371882197196	c_{46}	44854326288
c_5	62058	c_{19}	1151871871548	c_{33}	36654953745364	c_{47}	17645803824
c_6	526939	c_{20}	1442495328054	c_{34}	−24197928265284	c_{48}	7405762592
c_7	2690814	c_{21}	−887053312368	c_{35}	−46267383326688	c_{49}	1320738912
c_8	−1449897	c_{22}	−6823052001996	c_{36}	−26237073413784	c_{50}	273649344
c_9	−39750052	c_{23}	−9114754689990	c_{37}	6651637967208	c_{51}	94846400
c_{10}	−197448186	c_{24}	528017030132	c_{38}	17458275373896	c_{52}	8774784
c_{11}	−107044647	c_{25}	22918101813552	c_{39}	10405491226968	c_{53}	1057536
c_{12}	1304494737	c_{26}	32394375014568	c_{40}	−185213536824	c_{54}	373248
c_{13}	6446959536	c_{27}	8810442614448	c_{41}	−3340012771608		
c_{14}	6027624726	c_{28}	−41763292316760	c_{42}	−2091938831272		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54 + 4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

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